Digital Twins as Material Passports:

Enabling Circular Economy and Embodied Carbon Reduction

in the Built Environment through Component Reuse

Seminar Paper

Vienna University of Technology | Digital Architecture and Planning

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May 2026

Abstract

Buildings and construction are responsible for roughly 37% of global CO2 emissions, a large and structurally underaddressed share of which comes from embodied carbon, the emissions locked into building materials during extraction, manufacturing, transport, and end-of-life processing (IEA, 2022). Unlike operational carbon, embodied carbon cannot be undone once it has been incurred. Installing solar panels or upgrading insulation decades later does nothing to reduce the carbon that was released when the steel was smelted or the concrete was poured. As operational carbon falls under policy pressure and grid decarbonisation, embodied carbon is on track to account for roughly 50% of the total lifetime carbon footprint of all new buildings constructed between now and 2050 (WorldGBC, 2019).

Direct component reuse is the most powerful tool available for reducing embodied carbon, because it avoids generating new upfront carbon altogether rather than merely cutting it. Yet actual structural reuse rates across the built environment remain critically low. The underlying reason is an information and trust problem. Without documented load histories, certified material data, and recorded condition information, structural components simply cannot be approved for reuse by engineers and structural designers.

This paper builds on existing work in Digital Twin and Material Passport research to propose a conceptual framework addressing this gap. The central idea is that a Digital Twin, functioning as a continuously updated Material Passport, can resolve this information deficit by accumulating component-level data across the entire building lifecycle. Importantly, the framework does not require dedicated new infrastructure: it leverages sensor and data systems that buildings are already expected to deploy for operational purposes, energy management, predictive maintenance, and the like, and redirects this same infrastructure toward enabling circular construction and embodied carbon savings. RFID-tagged prefabricated components are tracked from installation through decades of service, with their structural history, load data, and condition records continuously updated in the Digital Twin. At end-of-life, this accumulated record enables informed, certified reuse decisions, turning structural components from anonymous demolition waste into documented, market-ready assets.

The argument unfolds through a structured lifecycle analysis using a single running example, an HEA 300 structural steel beam, and culminates in a direct comparison of end-of-life scenarios with and without a DT-based Material Passport. The discussion connects this framework to the broader industry shift toward prefabricated, design-for-disassembly construction and positions the Digital Twin as the informational layer that makes circular construction viable at scale.

Keywords: Material Passport, Digital Twin, Embodied Carbon, Circular Economy, Life Cycle Assessment, Building Lifecycle

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1. Introduction

The built environment is one of the most carbon-intensive sectors in the world. Buildings and construction together account for nearly 37% of global energy-related CO2 emissions, according to the International Energy Agency (IEA, 2022). While policy and technology have driven real progress in cutting operational carbon, a parallel problem has received far less attention: the carbon embedded in building materials themselves.

Embodied carbon covers the greenhouse gas emissions tied to the full material chain, extraction, manufacturing, transport, construction, and end-of-life processing. What makes it fundamentally different from operational carbon is that it cannot be walked back. As Chen et al. (2021) point out, while operational carbon can be gradually reduced over time through renewables and better insulation, embodied carbon is fixed the moment a building is built. A tonne of CO2 emitted when structural steel is produced is in the atmosphere from that instant onwards, no amount of solar panels installed decades later changes that. This irreversibility gives embodied carbon particular urgency when near-term emissions reductions are what the climate requires.

The scale of the problem becomes clear when projected forward. The World Green Building Council (2019) estimates that embodied carbon from new construction will account for around 50% of the total lifetime carbon footprint of all new buildings built between now and 2050. This is not because embodied carbon is growing, it is because operational carbon is shrinking as energy codes tighten and electricity grids decarbonise, making the carbon locked in at the moment of construction proportionally more significant.

The scale of construction and demolition waste (CDW) adds a further dimension to the problem. CDW makes up at least 30% of all solid waste generated globally (Zhang et al., 2022; Banihashemi et al., 2024; estimates vary widely across methodologies and regions, with some reaching up to 60%), making construction the single largest contributor to the global waste stream. Yet despite headline recycling figures that look encouraging, direct component reuse, the only strategy that avoids generating new embodied carbon entirely, accounts for only a small fraction of end-of-life outcomes, as discussed in Section 2.2.

Direct structural reuse is the most powerful available strategy for reducing both embodied carbon and CDW (NIST, 2024). Yet despite broad acknowledgment of this potential, actual reuse rates remain critically low. Su et al. (2026) point to limited digitalisation across the waste disposal chain as the primary cause, noting that it prevents seamless information exchange and efficient material recovery. The Steel Construction Institute's own protocols (SCI, 2019) confirm that reuse without verified material data requires costly hardness testing on every single component, a requirement that makes reuse economically unattractive at any meaningful scale.

Two trends are converging to create a new opportunity. Research on Digital Twin technology in construction has grown dramatically. Su et al. (2024) document a tenfold increase in published work between 2018 and 2022, with growing adoption in building management and energy optimisation. At the same time, the construction industry is

shifting toward prefabricated, modular systems with reversible connections (Banihashemi et al., 2024), a market projected to grow from USD 146 billion in 2024 to USD 208 billion by 2030 (Arizton, 2025). Prefabricated components are ideal candidates for reuse, but only if the informational infrastructure exists to support it.

This paper argues that the Digital Twin, operating as a continuously updated Material Passport, is precisely that infrastructure.

The paper is structured as follows. Section 2 covers the foundational concepts: Life Cycle Assessment and embodied carbon accounting, the circular economy principles that motivate the framework, and the basic principles of a Digital Twin. Section 3 sets out the core problem and research question. Section 4 develops the framework across the four lifecycle phases. Section 5 discusses implications and limitations. Section 6 concludes. A single structural component, the I-beam, runs as a concrete example throughout, introduced in Section 2.1 and carried forward through each subsequent phase.

2. Foundations

2.1 Life Cycle Assessment and Embodied Carbon

Life Cycle Assessment (LCA) is the standard methodological framework for quantifying the environmental impacts of a building over its full lifespan. The most widely adopted standard for building LCA in Europe is BS EN 15978 (British Standards Institution, 2011), which defines a structured set of lifecycle stages summarised in Table 1.

To understand where embodied carbon actually comes from, and why it is so hard to reduce once it has been incurred, it helps to work through a concrete example. This paper uses a single structural component as a running illustration: an HEA 300 structural steel beam, 8.4 metres long, weighing 739 kg, manufactured from S355 primary steel. Table 1 shows both the lifecycle stage definitions and the embodied carbon values calculated for this specific beam. Full calculations are set out in Appendix A.

The production stage (A1–A3) dominates the embodied carbon profile of structural materials, capturing the energy-intensive processes of raw material extraction, transport to the mill, and manufacturing (NIST, 2024). For the HEA 300 beam, cradle-to-gate emissions total 1,145 kgCO2e, around 76% of the overall lifecycle footprint. This makes A1–A3 by far the largest lever for carbon reduction. The construction stage (A4–A5) is comparatively minor: 44 kgCO2e for transport (assumed route: ThyssenKrupp Steel in Duisburg to a construction site in Munich, 590 km by road) and 3 kgCO2e for crane operation and installation. The use stage (B1–B7) generates no further embodied carbon for structural steel, which typically needs no maintenance or replacement over a building's service life. At end-of-life (C1–C4), deconstruction, transport, and disposal each contribute only a few kgCO2e. The dominant share comes from waste processing (C3), where EAF (Electric Arc Furnace) melting of the scrap adds 296 kgCO2e, almost

the entire end-of-life footprint. Summed across all stages, the linear lifecycle of the HEA 300 beam results in a total embodied carbon of 1,497 kgCO2e, as shown in Figure 1.

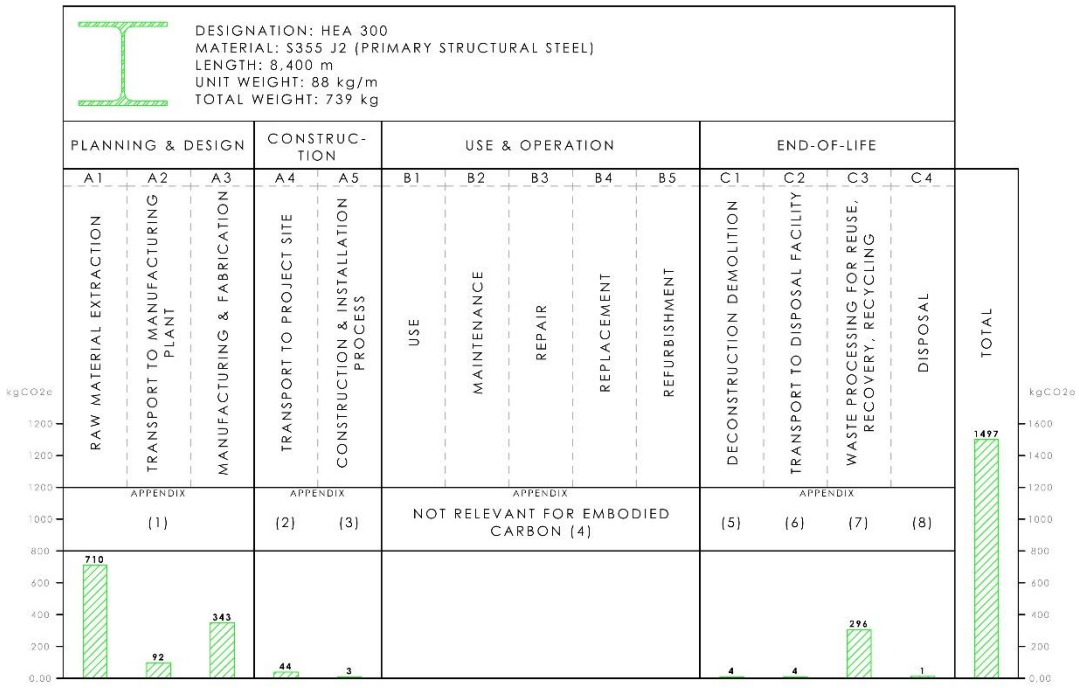


Figure 1. Embodied carbon contributions per EN 15978 lifecycle stage for the HEA 300 beam, linear scenario. All values in kgCO2e.

2.2 Circular Economy

The circular economy (CE) is a systemic framework aimed at decoupling economic activity from the consumption of finite resources and the production of waste. Rather than following the linear take-make-dispose model, CE principles seek to keep materials and products in use at their highest possible value for as long as possible (Keles et al., 2025). In construction, this means moving away from a paradigm in which buildings are demolished and their materials downgraded or landfilled, toward one in which buildings function as material banks, repositories of documented, recoverable components available for future use.

As noted in the introduction, construction and demolition waste accounts for around 30% of all solid waste generated globally (Zhang et al., 2022; Banihashemi et al., 2024). Despite recycling rates that look high on paper, the reality is considerably less impressive. Most CDW is downcycled rather than genuinely recovered: concrete is crushed into low-grade fill aggregate, structural steel is melted down in energy-intensive electric arc furnaces, a process that itself emits 400–800 kg CO_2 per tonne, and recast as lower-quality secondary products (World Steel Association, 2022; NIST, 2024). The label 'recycled' obscures the fact that most material is degraded and its embodied carbon partially re-emitted in the process. Direct component reuse, the only strategy that preserves full material value and avoids new embodied carbon entirely, currently accounts for only a very small fraction of end-of-life outcomes.

Table 2 illustrates the embodied carbon consequences of linear versus circular end-of-life pathways for the HEA 300 I-beam, showing that direct reuse reduces net lifecycle carbon by around 96% compared to a fully linear scenario without recycling, and by around 93% compared to a recycling scenario involving EAF melting with a secondary steel credit.

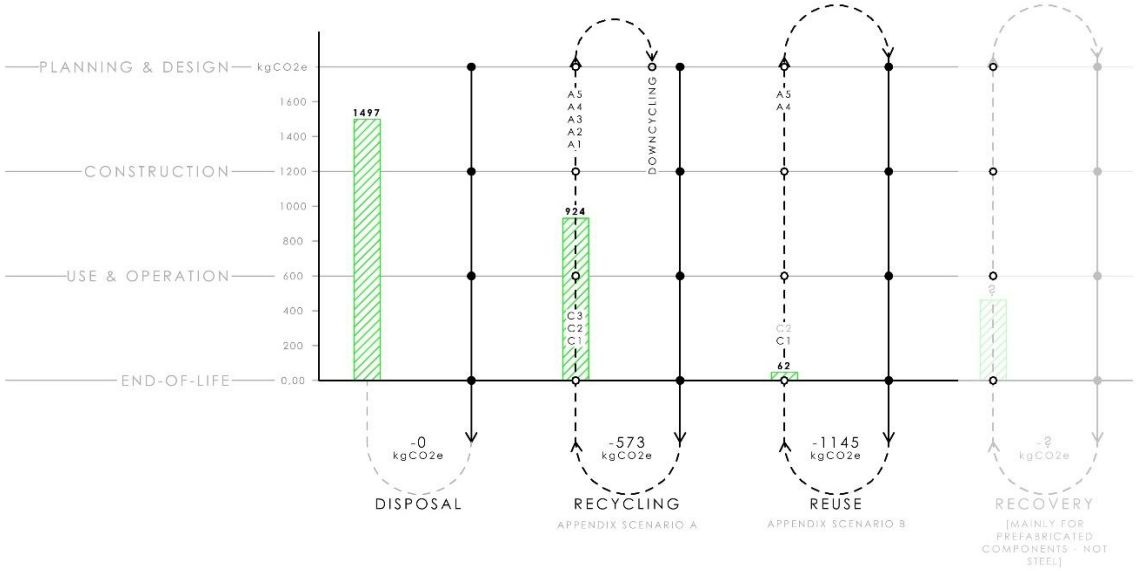


Figure 2. Embodied carbon of the HEA 300 beam across different end-of-life pathways, in kgCO2e. Own calculations, see Appendix A for methodology and emission factors.

The difference is substantial. Scaled to a small to medium-sized office building of around 5,000 m² with approximately 450 tonnes of structural steel, substituting direct reuse for linear demolition would avoid roughly 526 tCO2e per lifecycle, equivalent to the annual carbon sequestration of a forest covering 55 to 110 football pitches. And this figure covers only the structural steel. Including further reusable components such as prefabricated wall panels, floor elements, and facade modules would amplify the effect considerably. At the scale of the construction sector as a whole, the implications are far-reaching.

Design for Disassembly (DfD) is the design philosophy that makes component reuse physically possible, by ensuring building elements can be separated cleanly at end-of-life. A critical prerequisite is the use of reversible connections. As Banihashemi et al. (2024) note, if connections are welded, glued, or cast in place rather than bolted or screwed, components are damaged or destroyed during disassembly and become unusable for reuse. The shift toward bolted, screwed, or slotted joints is therefore not simply a technical preference, it is a fundamental design requirement for circular material flows.

This DfD imperative is converging with a broader industry shift toward prefabricated construction. Prefabricated components, structural elements, facade panels, wall cassettes, floor modules, offer faster build times, better quality control, reduced on-site waste, and standardised geometries that make disassembly and reuse more straightforward (Banihashemi et al., 2024). The NIST (2024) report and Keles et al. (2025) both identify prefabrication as an effective route to reducing embodied carbon through improved material efficiency and end-of-life component recovery.

2.3 Digital Twin

Building Information Modelling (BIM) has been the dominant digital infrastructure of the construction industry for the past two decades. A BIM model is a semantically rich, three-dimensional digital representation of a building containing geometric, material, and systems data, and serves as a shared information environment across design and construction phases (Cespedes-Cubides & Jradi, 2024).

The Digital Twin takes BIM a significant step further. As Su et al. (2024) describe, BIM is primarily focused on design and construction and does not typically involve real-time data analysis during operation. A Digital Twin, by contrast, is defined by a bidirectional, real-time coupling between the physical asset and its virtual representation. Sensor data from the physical building continuously updates the virtual model, while analyses and simulations run in the virtual model feed back into decisions about the physical asset.

The practical foundation of a Digital Twin is the integration of BIM with IoT technologies. As Cespedes-Cubides & Jradi (2024) describe, IoT systems, sensors, data pipelines, and processing infrastructure, extend BIM models into Digital Twins that reflect the actual behaviour and properties of the building throughout its lifecycle. The research field has grown dramatically: Su et al. (2024) document published work rising from 37 papers across the period 1973–2015 to 2,989 papers in 2021 alone. Despite this growth, the literature has focused overwhelmingly on construction, operation, and maintenance phases, including specific applications such as energy performance evaluation in existing buildings (Kaewunruen et al., 2018). The demolition and recovery phase has remained largely neglected, despite being arguably the most critical for circular economy outcomes (Su et al., 2024; Opoku et al., 2021). The framework presented here places end-of-life deliberately at its centre.

Table 1 maps Digital Twin functions across the full building lifecycle. While end-of-life is the primary focus of this framework, it does not operate in isolation: the same digital infrastructure that enables the Material Passport simultaneously delivers value across all lifecycle phases, from design through operation. Table 1 illustrates a selection of these functions.

PLANNING & DESIGN	CONSTRUCTION	USE & OPERATION	END-OF-LIFE
PARAMETRIC DESIGN OPTIMIZATION	CONSTRUCTION PROGRESS MONITORING	ENERGY MANAGEMENT	INFORMED DECONSTRUCTION PLANNING
DATA-INFORMED MATERIAL ANALYSIS	AS-BUILT DOCUMENTATION	PREDICTIVE MAINTENANCE	COMPONENT REUSE CERTIFICATION
PERFORMANCE SIMULATION	TRACKING VIA RFID	STRUCTURAL HEALTH MONITORING	MATERIAL BANK MATCHING
EMBODIED CARBON ESTIMATION	QUALITY CONTROL & INSPECTION	MATERIAL PASSPORT UPDATES	TARGETED COMPONENT REFURBISHMENT
	SITE LOGISTICS OPTIMIZATION	INDOOR ENVIRONMENTAL MONITORING	
		OCCUPANCY ANALYSIS	

Table 1. Selected Digital Twin functions across the four building lifecycle stages. Own representation, adapted from Opoku et al. (2021), Banihashemi et al. (2024), Cespedes-Cubides & Jradi (2024), and Su et al. (2024, 2026).

3. The Problem and Research Question

The case for component reuse as an embodied carbon reduction strategy is well established. Yet actual direct reuse rates remain critically low across the built environment. Even for structural steel, the material stream with the most developed reuse protocols, direct reuse accounts for only a small fraction of total recovery, with the vast majority (86%) going to EAF recycling rather than component reuse (Su et al., 2026; SCI, 2019). For all other major building material categories, concrete, glazing systems, prefabricated composites, insulation, direct component reuse is effectively negligible. Comprehensive cross-material reuse data does not exist in the literature, which is itself symptomatic of the problem: without systematic documentation of what is built and how, what can be reused cannot be tracked.

Su et al. (2026) identify limited digitalisation across the waste disposal chain as the primary cause, noting that it prevents seamless information exchange and efficient circulation of building demolition waste. Banihashemi et al. (2024) confirm that most building components end up as CDW precisely because buildings today do not have digitised material passports. As Mao and Cao (2025) establish in their systematic review of material passport implementations, the absence of standardised and trusted data at the decision-making level is the central barrier to circularity, and this is exactly the information and trust problem this framework is designed to address.

The root cause of low reuse rates is not a lack of intent. It is a structural information and trust deficit. When a structural engineer is asked to approve a reclaimed I-beam for use in a new building, the problem is straightforward: without documented evidence of the beam's material grade, load history, maintenance record, and current condition, a responsible certification of structural adequacy is hardly possible (Mao & Cao, 2025). The consequences of an error are catastrophic, structural failure. There is no room for assumption.

The Steel Construction Institute's protocol for structural steel reuse (SCI P427, 2019) puts formal shape to this problem. The protocol mandates non-destructive testing, specifically a hardness test on every individual reclaimed structural component and explicitly calls for a national database for new structural steelwork in which all mechanical and chemical properties are securely stored. With that provenance established, steel can be reused with confidence, free from the testing burden and conservative assumptions that currently make reuse impractical at scale (SCI, 2019). This proposed database is, in conceptual terms, precisely what a Digital Twin-based Material Passport would provide, not only for structural steel, but for every tracked component across all material types.

Beyond the certification barrier, two further obstacles compound the problem. First, there is a coordination failure: without a mechanism to match available reclaimed components with projects that need them, the market for structural reuse cannot function. Second, there is a deconstruction cost asymmetry: careful, selective deconstruction costs significantly

more than conventional demolition. Without the informational infrastructure to monetise the value of reusable components, the economic incentive to invest in it does not exist.

A third barrier is specific to prefabricated components, a category whose share of total construction is growing rapidly (Banihashemi et al., 2024; Arizton, 2025). As the industry increasingly adopts prefabricated wall panels, floor cassettes, and facade elements, a new challenge emerges: these components may incorporate insulation or embedded systems whose condition after decades of service is not visible from the exterior. A wall panel exposed to elevated humidity for five years may have compromised insulation performance, but without sensor data from the use phase, this cannot be assessed without invasive inspection. The Material Passport addresses this directly: by retaining condition data from the use phase, it enables targeted, evidence-based assessment of each component, making it possible to determine what can be reused directly, what requires inspection or partial replacement, and what can be certified without further testing.

These barriers, the information and trust deficit, the coordination failure, the deconstruction cost asymmetry, and the hidden condition problem, share a common root: the absence of a trusted, comprehensive, component-level information record that persists across the building lifecycle. Against this background, the paper poses the following research question:

> *How can a Digital Twin-based Material Passport, continuously maintained across the full building lifecycle, resolve the information and trust deficit that currently prevents the direct reuse or informed recovery of building components at end-of-life and what does an implementation framework for such a system look like?*

4. Framework: The DT-Enabled Material Passport across the Building Lifecycle

The framework developed here conceptualises the Digital Twin as the informational backbone of a building's material lifecycle. The Material Passport is not a standalone system but an integral part of the Digital Twin itself, the layer that makes circular material flows structurally possible (Keles et al., 2025). Before tracing the framework phase by phase, it helps to understand the core technological architecture and how its three constituent layers interact.

The framework operates across three interconnected levels. The base is the physical layer: the actual building components, structural steel members, prefabricated wall panels, facade elements, floor modules, each tagged with a passive RFID chip during manufacturing or on delivery to site (Banihashemi et al., 2024). The chip carries a unique identifier, giving each component a permanent physical identity that persists for the life of the building and survives demolition. The RFID tag does not store the passport data itself. It serves as the physical key that links the component to its complete digital record, accessible to any authorised party with a standard RFID reader at any point in the component's life, or equally by selecting the corresponding element directly within the Digital Twin platform.

The second level is the Digital Twin platform. Built on a BIM geometry model and continuously enriched with IoT sensor data (Cespedes-Cubides & Jradi, 2024; Opoku et al., 2021), it serves as the central hub where all component-level data is aggregated, interpreted, and stored over the building's lifetime. Each physical component is represented in the Digital Twin both geometrically, as a precisely located element in the virtual model, and informationally, through its associated Material Passport. The Digital Twin is therefore not a static 3D model. It is a living, continuously updated information system that knows where every tracked component is, what it is made of, and what it has experienced throughout its service life.

The third level is the Material Passport itself. As part of the Digital Twin, it constitutes the structured, component-level record for each building element (Mao & Cao, 2025). It is simultaneously accessible via the component's RFID tag and visible within the Digital Twin platform, ensuring that the component and its record remain permanently linked regardless of how many times the building changes hands or how much time passes.

A key feature of this architecture is how the Material Passport is populated during the use phase. Rather than requiring a dedicated sensing infrastructure built purely for passport purposes, the framework draws on data already generated by the building's operational systems (Cespedes-Cubides & Jradi, 2024; Wang et al., 2022). IoT sensors deployed for energy management, indoor climate monitoring, and facility management continuously feed data into the Digital Twin, which evaluates these streams and assigns relevant readings to the appropriate components based on location and data type. Temperature and humidity readings are routed to facade panels, load measurements to structural members, air quality data to interior partitions. As a result, each component's Material Passport grows passively and continuously throughout the use phase, as a byproduct of normal

building operation, without manual intervention. Exceptions apply where significant changes occur, such as component replacement, renovation, or structural modifications, which require manual updates to the relevant passport entries.

The data architecture underlying this system calls for a flexible, document-oriented database rather than a conventional relational one. Because different component types carry different data fields and file attachments, an I-beam requires a structural calculation report and mill certificate, a prefabricated wall panel requires a building physics assessment and fire safety certificate, a NoSQL database such as MongoDB is the appropriate choice. MongoDB stores each component record as a flexible JSON document that can hold any combination of structured data fields and file attachments, including PDFs, CAD/IFC files, and sensor time series, without requiring a fixed schema (Chen et al., 2021).

The framework is organised around the four lifecycle phases defined in EN 15978 and is illustrated throughout using the running I-beam example introduced in Section 2.1, shown in Figure 3.

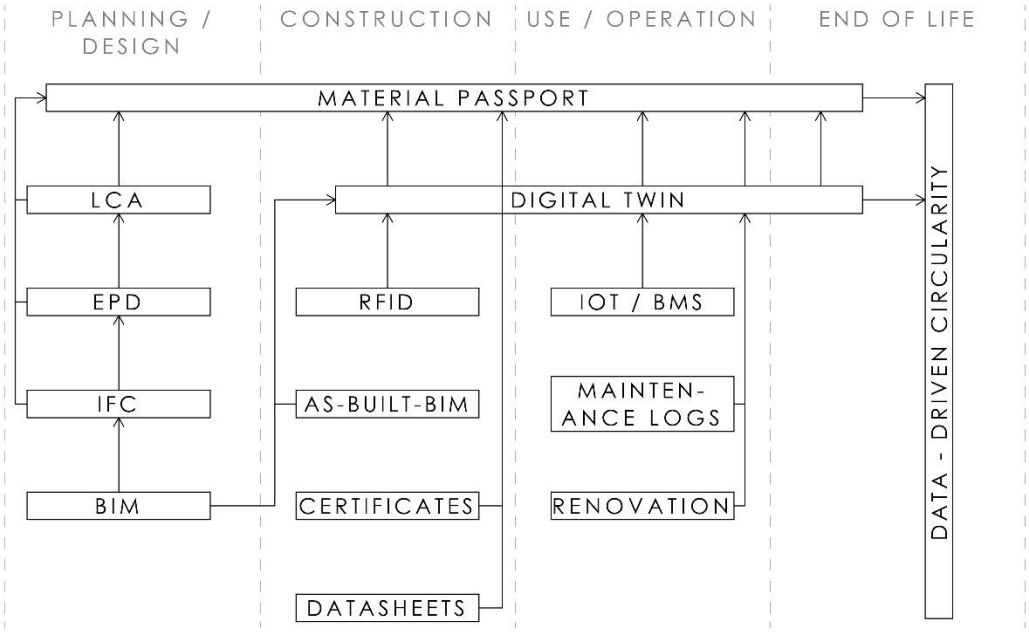


Figure 3. Data flows into the DT-enabled Material Passport across the four building lifecycle stages.

4.1 Design Phase

The design phase is the most consequential moment in a building's embodied carbon lifecycle, not because emissions are incurred here, but because the decisions made here determine all future embodied carbon outcomes (Chen et al., 2021). The choice of structural system, material specification, and connection detailing all fix the carbon trajectory of the building decades before demolition.

In this framework, the design phase is where the Material Passport is formally established. Once the design is complete, an IFC export is generated from the BIM model, from which the embodied carbon for each component is automatically calculated based on material quantities and ECF (Embodied Carbon Factor) values from established databases such as the ICE Database. Tools like OneClick LCA and SimaPro already perform LCA calculations from IFC exports at the building level. The framework extends this logic to the component level, so that the resulting A1–A3 embodied carbon values are automatically written to each component's Material Passport in the Digital Twin (Chen et al., 2021).

As the design develops, further data is added to the passport: manufacturer datasheets, EPDs, structural reports, fire safety certificates, CAD and IFC files of individual components, building physics assessments, and more (Banihashemi et al., 2024; Mao & Cao, 2025). As Banihashemi et al. (2024) note, material passports are created at the design stage and updated throughout the building's lifecycle. Because these documents span different file types and data structures, MongoDB, as described above, is the appropriate storage solution. AI-assisted document extraction could additionally be used to identify relevant parameters from uploaded files automatically and assign them to the correct passport fields, further reducing manual input.

At the end of section 4.4 an example of the complete data structure of the Material Passport is illustrated, using the running I-beam example (Figure 5).

4.2 Construction Phase

The construction phase is when each physical component and its Material Passport are permanently linked. Passive RFID chips are embedded in or attached to each tracked component during manufacturing or before delivery to site. These chips are battery-free, have a virtually unlimited lifespan, and cost as little as EUR 0.10 to 2.00 per tag. They permanently connect each physical element to its digital record in the Digital Twin, as described in Section 4 (Su et al., 2024; Banihashemi et al., 2024).

During construction, the Digital Twin is updated to reflect as-built conditions. Design models frequently diverge from what is actually built due to on-site modifications, component substitutions, or installation variations. The as-built BIM model, updated through site surveys, RFID scanning, and contractor inputs, becomes the definitive starting point for the use-phase passport.

With as-built data confirmed, the A1–A3 embodied carbon values estimated at design stage can be updated and refined. Now that the actual manufacturer, batch, and delivery distance are known, the LCA calculation is revised accordingly, replacing any generic estimates with verified real-world values (Chen et al., 2021).

The construction phase adds the following key entries to the passport: RFID tag unique identifier, installation date, exact as-built position in the building, actual manufacturer and batch number, and transport distance with resulting A4 and A5 carbon values. Most critically, component-specific certification documents are attached as PDFs. For example, the mill certificate for structural steel confirming material grade and chemical composition, or equivalent manufacturer certificates for prefabricated components. These are precisely the documents a structural engineer will need decades later when assessing the component for reuse.

4.3 Use Phase

The use phase is the longest in a building's lifecycle, typically spanning 50 to 80 years. While structurally little changes during this time, it is precisely this phase in which the Material Passport is continuously enriched: IoT sensors and building management systems feed operational data into the Digital Twin, progressively building the evidence base that will enable informed reuse decisions at end-of-life.

As described in the framework architecture above, new buildings are increasingly being equipped with IoT sensor networks for operational purposes, energy management, climate monitoring, facility management, supported by machine-learning-based prediction and diagnostics (Cespedes-Cubides & Jradi, 2024; Wang et al., 2022). The data streams generated by these systems are systematically assigned to the corresponding component records in the Digital Twin, based on component location and data relevance. For example, humidity readings are particularly important for prefabricated facade panels and insulated wall elements, where moisture ingress can compromise insulation performance over time. On the other hand, UV intensity and temperature fluctuation data are most relevant for glazing systems and window elements, where they accelerate the degradation of seals and coatings.

Over decades, this accumulation produces a detailed, timestamped operational biography for every tracked component. Sensor-generated data such as environmental exposure readings flows automatically into the passport, while maintenance and repair events are logged manually by facility management against the corresponding component ID. For structurally critical components or safety-relevant structures such as bridges, additional targeted sensors, strain gauges, for instance, can be deployed where the application warrants it (Banihashemi et al., 2024).

Figure 4 illustrates how the three layers, the physical building, the Digital Twin, and the Material Passport, interact during the use phase to continuously capture, classify, and store component-level data.

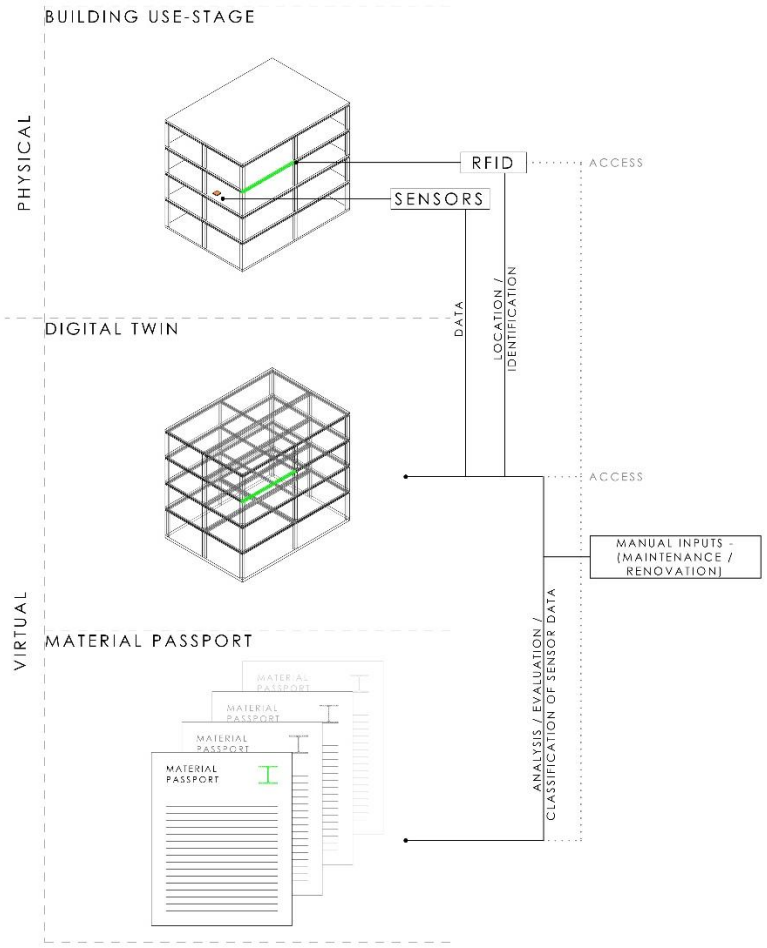


Figure 4. Interaction of the physical building, the Digital Twin, and the Material Passport during the use phase

4.4 End-of-Life

The end-of-life stage is the moment at which the value of the accumulated Material Passport is realised. The following two paragraphs contrast the current state of practice in a largely linear economy, where most building components are demolished, downcycled, or landfilled. On the other hand the DT-enabled Material Passport approach proposed here, which aims to enable circular material flows through direct component reuse.

Without component-level documentation, there is no basis on which to assess whether selective deconstruction is feasible. Manufacturers of prefabricated components cannot evaluate whether elements are worth returning for refurbishment and therefore do not accept them. Structural components lose their provenance entirely once conventional demolition begins. Across all material categories, the default pathway is anonymous scrap, downcycling, or landfill, driven not by an absence of intent to reuse, but by the absence of the informational infrastructure that would make reuse possible (Su et al., 2026; Banihashemi et al., 2024). The barriers underlying this outcome are discussed in detail in Section 3.

The DT-enabled Material Passport inverts this situation entirely. Before a single element is touched, the Digital Twin provides a complete component inventory, locations, connection types, load histories, and passport completeness for every tracked element, enabling a structured deconstruction plan sequenced by reuse value rather than demolition logistics. For structural components, a complete passport with no anomalous load events and a DfD-compatible connection type allows direct engineering approval without the hardness testing that currently makes reclaimed steel economically unviable at scale (SCI, 2019). For prefabricated components, manufacturers can direct their refurbishment precisely: a wall panel with a documented humidity event in year fifteen receives a targeted inspection of its insulation layer only, while its structural frame is approved for direct reuse, eliminating both blanket refurbishment costs and liability gaps. At the system level, buildings equipped with DT Material Passports become material banks. Their components are visible, documented, and matchable to incoming construction projects before deconstruction even begins. This vision was originally articulated in the BAMB (Buildings as Material Banks) framework (BAMB, 2016) and is increasingly enabled by recent advances in DT-supported reverse supply chains (Banihashemi et al., 2024; Su et al., 2026).

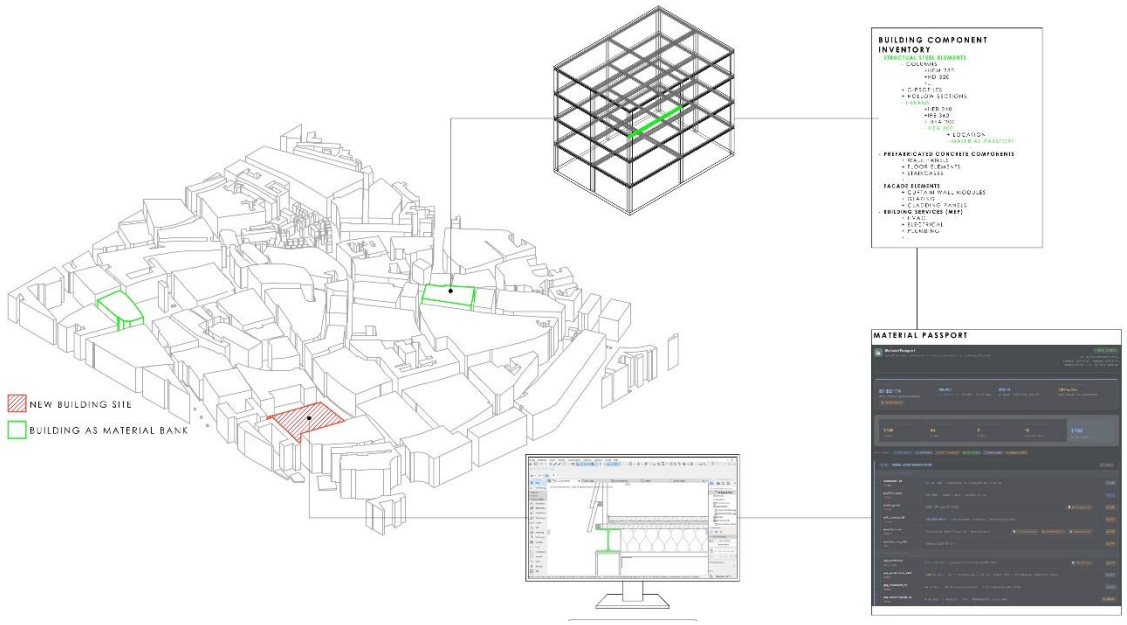


Figure 5. End-of-life buildings as queryable material banks: Material Passport-enabled component reuse in new construction projects (similar to the concept introduced by BAMB, 2016).

The QR code or link below shows an interactive example of the consolidated Material Passport for the HEA 300 I-beam discussed throughout this paper, demonstrating how the data accumulated across all four lifecycle phases comes together in a single component-level record. The content of the passport, including all values, documents, and the underlying data structure, was developed by the author. The HTML implementation and visual layout were generated with the assistance of Anthropic's Claude (Anthropic, 2025).



Link: Material Passport, HEA 300 I-Beam

Link: https://adriandfr.github.io/material-passport-demo/

The QR-Code / Link as an example how the RFID reader or the DT accesses the Material Passport. | Interactive example of the Material Passport

5. Limitations

This paper presents a conceptual framework and does not empirically validate the costs or implementation dynamics of DT-based Material Passport systems. Several important limitations need to be acknowledged.

A first practical limitation is the number of stakeholders involved. A DT-based Material Passport requires sustained coordination across design teams, construction contractors, manufacturers, successive facility managers, maintenance providers, and eventually deconstruction contractors, often over a horizon of more than fifty years. Without agreed open data schemas, clearly defined ownership of the passport record, and interoperable BIM standards, the documentation chain is at risk of fragmentation at every handover, which would undermine the entire framework (Banihashemi et al., 2024).

A second limitation concerns whether direct reuse remains environmentally rational under all future conditions. The carbon advantage of reuse over recycling weakens as electricity grids decarbonise and steel production runs increasingly on renewables. Two arguments preserve the relevance of the framework regardless. First, steel quality continues to degrade through tramp element contamination across recycling cycles, independent of energy source. Second, keeping materials circulating within the European economy reduces structural dependency on imported raw materials, a strategic priority explicitly pursued by the EU (Eurostat, 2024).

A third limitation is the existing building stock. The framework assumes a Digital Twin established at the design stage, while most buildings standing today predate the digital era (Cespedes-Cubides & Jradi, 2024; Lu et al., 2020). Retrofitting through Scan-to-BIM, AI-based component recognition, and retrospective RFID tagging during maintenance is

technically possible, but will always produce a less complete passport than a prospective one built from the outset (Lu et al., 2020). Nevertheless, at scale, the aggregated passport data would constitute a uniquely granular, component-level resource for the broader industry, enabling AI-assisted stock assessments (including Scan-to-BIM optimisation), embodied carbon benchmarking, and material flow analyses across the built environment.

Finally, the broader economic viability of the framework remains an open question. Whether a fully circular construction economy built around DT Material Passports is profitable in aggregate is a separate question this paper does not resolve. The economic potential is nonetheless worth noting. Based on rough market estimates, a certified, passport-documented structural steel component available for direct reuse commands approximately 1.5 to 2 times the value it would yield as anonymous EAF scrap. For prefabricated components, where the value lies not just in the raw material but in the manufactured assembly itself, this contrast is likely far more pronounced. Scaled across all recoverable components in a building, the aggregate economic potential is substantial. The point here is simply that the economic case for DT-enabled reuse is not implausible, even if a precise quantification lies outside the scope of this work. Market readiness remains the more immediate challenge: functional secondary markets for certified reclaimed components are still nascent, and the reverse supply chain infrastructure required to channel them efficiently remains underdeveloped (Su et al., 2026). In the near term, the framework is therefore most applicable to large-scale commercial and industrial projects where component values are high and deconstruction is planned well in advance.

6. Conclusion

This paper has argued that the Digital Twin, operating as a continuously updated Material Passport, provides the informational infrastructure needed to resolve the trust and documentation deficit that currently prevents direct reuse of structural building components at end-of-life. The framework draws on sensor and data systems that modern buildings will increasingly deploy for operational purposes, generating the Material Passport as a byproduct of normal building management rather than requiring dedicated new infrastructure.

At end-of-life, the accumulated passport enables informed deconstruction, passport-guided engineering approval without individual material testing, and targeted manufacturer refurbishment of returning prefabricated components. The result is a shift from anonymous demolition waste to documented, market-ready material assets.

The broader context reinforces this direction. Emerging EU regulation, from the Taxonomy Regulation and the Corporate Sustainability Reporting Directive to the forthcoming Digital Product Passport under the Ecodesign for Sustainable Products Regulation, is pushing the construction sector toward exactly the kind of mandatory component-level lifecycle documentation that a DT-based Material Passport delivers as an inherent output (EU 2020/852; Directive (EU) 2022/2464; EU 2024/1781). Keeping materials circulating within the European economy also directly supports the EU's stated objective of reducing structural dependency on imported raw materials.

Taken together, the framework points toward a construction industry in which buildings are no longer terminal products but long-lived, documented material reservoirs, with components that retain their identity, value, and reusability across successive lifecycles.

Appendix A: Key Calculations

A.1 HEA 300 I-Beam, Embodied Carbon (Static LCA, Chen et al. 2021 framework)
Calculations follow the framework proposed by Chen et al. (2021), using ECF values from the Inventory of Carbon and Energy database (Hammond & Jones, University of Bath, Version 3.0) and emission factors from EN 15978 and EN 16258.
Component: HEA 300, L = 8.4 m, unit weight = 88 kg/m, total weight = 739 kg, S355 J2.

(1) A1–A3: Production stage (cradle-to-gate)
Aggregated embodied carbon calculated as E_pro = Q × ECF (Chen et al., 2021, Eq. 1), with ECF = 1.55 kgCO2e/kg for primary structural steel (ICE Database, Hammond & Jones, University of Bath, Version 3.0).

- E_pro = 739 kg × 1.55 kgCO2e/kg = **1,145 kgCO2e**

The aggregated A1–A3 value is disaggregated into individual stages using typical distribution ratios reported in the literature on primary steel production (Hammond & Jones, 2019; World Steel Association, 2022): A1 ≈ 62%, A2 ≈ 8%, A3 ≈ 30%.
A1: Raw material supply (iron ore, coking coal, limestone)

- E_A1 = 0.62 × 1,145 = **710 kgCO2e**

A2: Transport to manufacturer (raw inputs to steel mill)

- E_A2 = 0.08 × 1,145 = **92 kgCO2e**

A3: Manufacturing (blast furnace, basic oxygen furnace, hot rolling)

- E_A3 = 0.30 × 1,145 = **343 kgCO2e**

(2) A4: Transport to construction site
Transport carbon calculated using EN 16258 road freight emission factor:

- E_A4 = m × d × EF_transport

where m = mass in tonnes, d = distance in km, EF_transport = 0.10 kgCO2e/tonne·km (EN 16258, road freight, average loaded truck). Assumed route: ThyssenKrupp Steel, Duisburg → construction site, Munich (590 km by road).

- E_A4 = 0.739 t × 590 km × 0.10 kgCO2e/tonne·km = **44 kgCO2e**

(3) A5: Construction / Installation

- E_A5 = m × EF_installation

where EF_installation = estimated 4 kgCO2e/tonne for crane operation and on-site equipment (RICS Whole Life Carbon Assessment, 2017).

- E_A5 = 0.739 t × 4 kgCO2e/t = **3 kgCO2e**

Total Upfront Carbon (A1–A5): 710 + 92 + 343 + 44 + 3 = 1,192 kgCO2e

(4) B1–B7: Use phase
Structural steel components do not emit further embodied carbon during the use phase under standard assumptions (no maintenance, no replacement). Operational energy (B6) is allocated at building level, not at component level.

- E_B1–B7 = **0 kgCO2e**

(5) C1: Deconstruction / Demolition

Energy and equipment for component-level removal from the building.

- E_C1 = m × EF_deconstruction

where EF_deconstruction = estimated 5 kgCO2e/tonne for conventional demolition equipment (RICS Whole Life Carbon Assessment, 2017).

- E_C1 = 0.739 t × 5 kgCO2e/t = **4 kgCO2e**

(6) C2: Transport to waste processing or reuse site

- E_C2 = m × d × EF_transport

Linear scenario: demolition site → scrap yard, assumed 50 km road freight.

- E_C2 = 0.739 t × 50 km × 0.10 kgCO2e/tonne·km = **4 kgCO2e**

(7) C3: Waste processing

EAF melting of scrap steel, secondary production process. ECF = approx. 0.40 kgCO2e/kg under EU grid mix 2025 (World Steel Association, 2022).

- E_C3 = 739 kg × 0.40 kgCO2e/kg = **296 kgCO2e**

(8) C4: Disposal

Slag waste only. Negligible for steel. Assumed EF_disposal = 1 kgCO2e/tonne.

- E_C4 = 0.739 t × 1 kgCO2e/t = **1 kgCO2e**

Total Lifecycle Embodied Carbon, Linear Scenario (A1–C4): 1,192 + 0 + 305 = 1,497 kgCO2e

A.2 End-of-Life Comparison: Recycling vs. Direct Reuse

The net carbon performance of the component depends on its end-of-life pathway. The comparison is made through Module D of EN 15978, which accounts for benefits and loads beyond the system boundary (avoided emissions from substituted primary production).

Scenario A: Linear (EAF Recycling)

The beam is removed, transported to a scrap yard, melted in an electric arc furnace, and recast as secondary steel. Due to tramp element contamination, recycled EAF steel is typically downcycled and cannot fully substitute primary steel. A 50% substitution credit is applied (Chen et al., 2021).

- E_total (A1–C4) = 1,497 kgCO2e
- Module D = -0.50 × E_pro = -0.50 × 1,145 = -573 kgCO2e
- **Net lifecycle carbon (Linear) = 1,497 - 573 = 924 kgCO2e**

Scenario B: Circular (Direct Reuse via DT Material Passport)

The beam is removed using selective deconstruction, transported regionally to its next installation site, inspected against passport data, and reinstalled without melting or new production. Module D credits the full avoided primary production of the beam this component substitutes.

C-stages in the reuse scenario:

- C1 Selective deconstruction (manual disassembly of bolted connections, slightly higher energy than wrecking demolition):
  - E_C1 = 0.739 t × 6 kgCO2e/t = **4 kgCO2e**
- C2 Regional transport to next site, assumed 86 km road freight:
  - E_C2 = 0.739 t × 86 km × 0.10 kgCO2e/tonne·km = **6 kgCO2e**
- C3 Inspection and passport-based certification (no waste processing required):
  - E_C3 = approx. **5 kgCO2e**
- C4 Disposal:
  - E_C4 = **0 kgCO2e**

Total C-stage emissions (Circular) = 4 + 6 + 5 + 0 = **15 kgCO2e**

Module D (avoided new production):

- Module D = -1.00 × E_pro = -1.00 × 1,145 = **-1,145 kgCO2e**

Net lifecycle carbon (Circular):

- E_total (A1–A5) + E_C1–C4 (Circular) + Module D
- = 1,192 + 15 − 1,145
- = **62 kgCO2e**

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